

Applications Involving Radian Measure

Arc Length

IMPORTANT INFO: Arc Length

Given a circle with radius r and a central angle θ , in radians, the length of the intercepted arc, denoted by s , is given by the following equation.

$$s = r\theta, \theta \text{ in radians}$$

As an alternative to a memorized rule like the one above, arc length can also be determined using ratio and proportion. For example, given a central angle θ in radians or degrees, we can construct a ratio of the following form.

$$\frac{\text{part}}{\text{whole}} = \frac{\theta}{360^\circ} \quad (\theta \text{ in degrees}) \quad - \text{OR} - \quad \frac{\text{part}}{\text{whole}} = \frac{\theta}{2\pi} \quad (\theta \text{ in radians})$$

We have a similar ratio for arc length.

$$\frac{\text{part}}{\text{whole}} = \frac{s}{2\pi r}$$

Since the arc length s , will be proportional to the central angle, which intercepts the arc, we can establish the following proportion statements.

$$\frac{\text{part}}{\text{whole}} = \frac{s}{2\pi r} = \frac{\theta}{360^\circ} \quad (\theta \text{ in degrees}) \quad - \text{OR} - \quad \frac{\text{part}}{\text{whole}} = \frac{s}{2\pi r} = \frac{\theta}{2\pi} \quad (\theta \text{ in radians})$$

Formulas like the one given above, $s = r\theta$, can be useful but oftentimes lead students to believe that mathematics is little more than a set of memorized rules and figures. This workbook will focus on comprehension of the underlying concepts. In the case of arc length, ratio and proportion may be a better approach.

EXAMPLE 1

Given an angle θ in radians, use ratio and proportion to verify the statement $s = r\theta$. Develop a similar statement for θ in degrees.

$$\begin{aligned} \frac{\text{part}}{\text{whole}} \quad \frac{\theta}{2\pi} &= \frac{s}{2\pi r} \\ s &= \cancel{2\pi} r \left(\frac{\theta}{\cancel{2\pi}} \right) \\ s &= r\theta \\ \theta &\text{ in radians} \end{aligned}$$

$$\begin{aligned} \frac{\text{part}}{\text{whole}} \quad \frac{\theta}{360^\circ} &= \frac{s}{2\pi r} \\ s &= 2\pi r \left(\frac{\theta}{360^\circ} \right) \\ s &= \frac{\pi}{180} r \theta \\ \theta &\text{ in degrees} \end{aligned}$$

EXAMPLE 2

Consider a circle with a diameter of 6 inches. Find the length of the arc intercepted by each central angle θ given below.

(a) $\theta = \frac{4\pi}{3}$

part
whole

$$\frac{4\pi/3}{2\pi} = \frac{s}{2\pi r}$$

$$s = 2\pi(3)\left(\frac{4\pi/3}{2\pi}\right)$$

$$s = 4\pi \text{ in.}$$

(b) $\theta = 225^\circ$

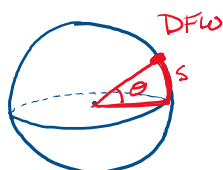
$$\frac{225^\circ}{360^\circ} = \frac{s}{2\pi r}$$

$$s = 2\pi(3)\left(\frac{225}{360}\right)$$

$$s = \frac{1350}{360}\pi = \frac{15}{4}\pi \text{ in.}$$

EXAMPLE 3

In an earlier lesson, we were given the Earth's circumference to be approximately 25,000 miles, with DFW International Airport located at approximately $32^\circ 53' 59''$ N, $97^\circ 2' 25''$ W. Roughly how far is a direct route flight from DFW to the Earth's equator? (Source: Google Maps, www.maps.google.com)



$$\frac{(32 + \frac{53}{60} + \frac{59}{360})^\circ}{360^\circ} \approx \frac{s}{25000}$$

$$s \approx 25000 \left(\frac{32.89972222}{360} \right)$$

$$s \approx 2284.70 \text{ mi}$$

One direct application for arc length is gearing. We interact with engineered systems that rely on gear ratios every day. For example, the vehicle you drive has various mechanical systems involving gearing. With each turn of your engine, there are multiple points where that rotational force is converted before power is delivered to the vehicle's wheels. A vehicle transmission is a complex system of gears with many effects, from low-end torque (power you'll need to get moving) to highway fuel economy. As a simplified example, the first gear on a vehicle may operate at an input:output ratio of 3.00:1. This means for every three rotations of the input gear (from the engine), the output gear (connected to the driveshaft) rotates once. As you move to higher gears, this ratio will change. Let's review an example of gear ratios.

EXAMPLE 4

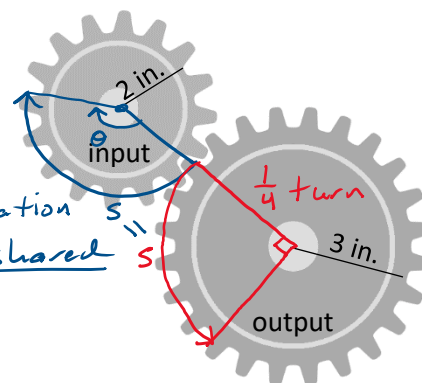
Using the gears shown here, answer the following.

- (a) How much must the input gear rotate to drive a one-quarter turn on the output gear?

Output $\frac{1}{4}$ turn $\rightarrow s = \frac{1}{4}(2\pi \cdot 3) = \frac{3\pi}{2}$ in.

We need the input gear to drive a $\frac{1}{4}$ rotation s for the output gear. This amounts to a shared arc length of $\frac{3\pi}{2}$ in.

Input $\frac{\theta}{360^\circ} = \frac{3\pi/2}{2\pi \cdot 2} \rightarrow \theta = 360^\circ \left(\frac{3\pi/2}{2\pi \cdot 2} \right)$
 $= 360^\circ \left(\frac{3}{8} \right) = 135^\circ$



- (b) What is the input:output ratio for this configuration?

$\frac{135}{360} : \frac{90}{360} \rightarrow \frac{135}{90} : \frac{90}{90} \rightarrow 1.5 : 1$

Every 1.5 rotations for the input gear drives 1 rotation on the output gear.

Linear and Angular Speed

Related to vehicle transmissions, a vehicle's engine provides us with an excellent example of our next topic, rotational (or *angular*) speed and straight-line (or *linear*) speed. As a vehicle's engine rotates, this angular speed is eventually converted at the wheels into linear speed. We summarize these concepts below.

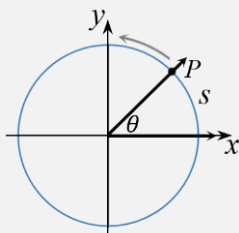
IMPORTANT INFO: Linear and Angular Speed

Consider a circle centered at the origin, with a central angle θ , and θ increasing at a constant rate. Let P be the point along the circle intercepted by the terminal side of θ .

The rate of change in the position of point P along the circle is the **linear speed**, denoted here as v .

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$v = \frac{s}{t}$$



The rate of change in the angle measure is the **angular speed**, denoted here as ω (omega).

$$\text{angular speed} = \frac{\text{amt. rotation}}{\text{time}}$$

$$\omega = \frac{\theta}{t}, \theta \text{ in radians}$$

It's likely that you're familiar with the above formula for linear speed. However, the equation $\omega = \theta/t$ is not so common and only applies to angles measured in radians. Rather than memorizing this formula, we will again leverage the principles of ratio and proportion to resolve problems involving linear and angular speed.

EXAMPLE 5

Consider a circle with a radius of 10 cm and central angle θ . Let P be the point along the circle intercepted by the terminal side of θ , with θ increasing at a constant rate of $\pi/5$ radians per second.

(a) What angle is formed by P after 1 second, 5 seconds, and 8 seconds?

$$\left(\frac{\pi}{5} \frac{\text{rad}}{\text{sec}}\right)(1 \text{ sec}) = \frac{\pi}{5} \text{ rad} \quad \left(\frac{\pi}{5} \frac{\text{rad}}{\text{sec}}\right)(5 \text{ sec}) = \pi \text{ rad.}$$

$$\left(\frac{\pi}{5} \frac{\text{rad}}{\text{sec}}\right)(8 \text{ sec}) = \frac{8\pi}{5} \text{ rad}$$

(b) Find the distance traveled by point P in 8 seconds.

$$\frac{8\pi/5}{2\pi} = \frac{s}{2\pi r} \rightarrow s = \cancel{2\pi} \cdot 10 \left(\frac{8\pi/5}{\cancel{2\pi}}\right) = \frac{80}{5}\pi = 16\pi \text{ cm.}$$

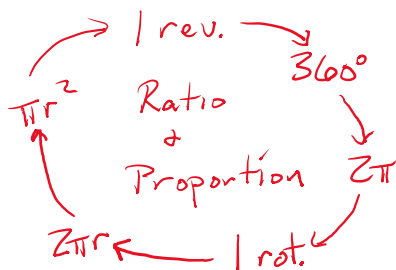
$\approx 50.27 \text{ cm.}$

(c) Find the linear speed at which point P is moving in centimeters per minute.

Convert $\frac{\text{rad}}{\text{sec}} \rightarrow \frac{\text{cm}}{\text{sec}}$ (rotational $\rightarrow$ linear)

$$\frac{\pi}{5} \left(\frac{\text{rad}}{\text{sec}}\right) \left(\frac{2\pi r \text{ cm}}{2\pi \text{ rad}}\right) \left(\frac{60 \text{ sec}}{1 \text{ min}}\right) = \frac{\pi}{5} (10 \text{ cm}) \left(\frac{60}{1 \text{ min}}\right)$$

$$= 120\pi \frac{\text{cm}}{\text{min}}$$



EXAMPLE 6

A wheel with a radius of 13 inches is rotating at 2 revolutions per second.

- (a) Find the angular speed for this wheel. Give your result in radians per second, degrees per second, and rotations per minute.

$$\frac{2 \cancel{\text{rev}}}{1 \text{ sec}} \left(\frac{2\pi \text{ rad}}{1 \cancel{\text{rev}}} \right) = 4\pi \frac{\text{rad}}{\text{sec}} \quad \frac{2 \cancel{\text{rev}}}{1 \text{ sec}} \left(\frac{360^\circ}{1 \cancel{\text{rev}}} \right) = 720 \frac{\text{deg}}{\text{sec}}$$

$$\frac{2 \cancel{\text{rev}}}{1 \text{ sec}} \left(\frac{1 \text{ rot}}{1 \cancel{\text{rev}}} \right) \left(\frac{60 \cancel{\text{sec}}}{1 \text{ min}} \right) = 120 \frac{\text{rot}}{\text{min}}$$

- (b) Find the linear speed for the wheel in miles per hour.

$$\begin{aligned} \frac{2 \cancel{\text{rev}}}{1 \text{ sec}} \left(\frac{2\pi r \text{ in}}{1 \cancel{\text{rev}}} \right) \left(\frac{1 \cancel{\text{ft}}}{12 \text{ in}} \right) \left(\frac{1 \cancel{\text{mi}}}{5280 \cancel{\text{ft}}} \right) \left(\frac{60 \cancel{\text{sec}}}{1 \text{ min}} \right) \left(\frac{60 \cancel{\text{min}}}{1 \text{ hr}} \right) \\ = \frac{2(2\pi \cdot 13)(60)(60)}{12(5280)} \frac{\text{mi}}{\text{hr}} \\ = \frac{187200\pi}{63360} \text{ mph} = \frac{65\pi}{22} \text{ mph} \approx 9.28 \text{ mph} \end{aligned}$$

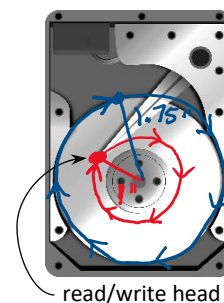
Diagram for linear speed calculation: A circle representing the wheel with radius r . The circumference is labeled $2\pi r$. One full rotation is labeled 1 rot. and 360° . One full revolution is labeled 1 rev. and 2π .

EXAMPLE 7

The hard disk for a computer has a diameter of 3.5 inches and a spin rate of 5400 RPM (rotations per minute).

- (a) What is the angular velocity of the hard disk expressed in radians per second?

$$\begin{aligned} \frac{5400 \cancel{\text{rot}}}{1 \cancel{\text{min}}} \left(\frac{2\pi \text{ rad}}{1 \cancel{\text{rot}}} \right) \left(\frac{1 \cancel{\text{min}}}{60 \text{ sec}} \right) &= \frac{5400(2\pi)}{60} \frac{\text{rad}}{\text{sec}} \\ &= 180\pi \frac{\text{rad}}{\text{sec}} \end{aligned}$$



- (b) If the read/write head of the disk drive is located 1 inch from the axis of rotation, what is the linear speed of the disk below it?

$$\begin{aligned} \frac{5400 \cancel{\text{rot}}}{1 \cancel{\text{min}}} \left(\frac{2\pi r \text{ in.}}{1 \cancel{\text{rot}}} \right) &= 10800\pi \cdot r \frac{\text{in}}{\text{min}} \\ &= 10800\pi \frac{\text{in}}{\text{min}} \end{aligned}$$

Diagram for linear speed calculation: A red arrow points to the radius r in the equation, with the label $r = 1$.

Area of a Sector

IMPORTANT INFO: Area of a Sector

Given a circle with radius r and a central angle θ in radians, the area of a sector A , is given by the following equation.

$$A = \frac{1}{2} r^2 \theta, \theta \text{ in radians}$$

As an alternative, the area of a sector can also be determined by ratio and proportion. Given a central angle θ in radians or degrees, we can construct a ratio of the following form.

$$\frac{\text{part}}{\text{whole}} = \frac{\theta}{360^\circ} \quad (\theta \text{ in degrees}) \quad - \text{OR} - \quad \frac{\text{part}}{\text{whole}} = \frac{\theta}{2\pi} \quad (\theta \text{ in radians})$$

We have a similar ratio for the area of a sector.

$$\frac{\text{part}}{\text{whole}} = \frac{A}{\pi r^2}$$

Since the area of a sector A , is proportional to the central angle which bounds the sector, we can establish the following proportion statements.

$$\frac{\text{part}}{\text{whole}} = \frac{A}{\pi r^2} = \frac{\theta}{360^\circ} \quad (\theta \text{ in degrees}) \quad - \text{OR} - \quad \frac{\text{part}}{\text{whole}} = \frac{A}{\pi r^2} = \frac{\theta}{2\pi} \quad (\theta \text{ in radians})$$

As mentioned before, memorized formulas like the one given above, $A = \frac{1}{2} r^2 \theta$, can be useful, but this workbook is concerned with comprehending the underlying concepts. We will again favor the approach of ratio and proportion over a memorized rule.

EXAMPLE 8

A building has revolving doors at a right-angled corner entry, as shown. The area swept out by the doors is 16π square feet, and the main landing has an area of 117π square feet. If city codes require steps to have a minimum tread depth of 12 inches, what is the total surface area of the step leading to the landing?

$$\frac{1}{4} \text{ door area} = \frac{1}{4} (16\pi) = 4\pi \text{ ft}^2$$

$$\text{landing} + \frac{1}{4} \text{ door} = 117\pi + 4\pi = 121\pi$$

$$\frac{121\pi}{\pi r_1^2} = \frac{90^\circ}{360^\circ} \rightarrow r_1^2 = 4(121)$$

$$r_1^2 = 484$$

$$r_1 = 22 \text{ ft}$$

Area of Step

$$\frac{1}{4} \pi (r_2)^2 - \frac{1}{4} \pi (r_1)^2$$

$$\frac{1}{4} \pi (22+1)^2 - \frac{1}{4} \pi (22)^2$$

$$\frac{529}{4} \pi - \frac{484}{4} \pi = \frac{45}{4} \pi \text{ ft}^2$$

